

Machine Learning Model for use in Grocery Price Prediction

Abstract-The goal of this project is to develop deep learning based forecasting models for grocery produce demand prediction using publicly available retail sales data to improve forecasting accuracy for perishable grocery times while reducing inventory waste and improving holiday demand prediction.

- I. Introduction
- II. Objectives
- III. Theory
- IV. Methodology
 1. MLP
 2. GRU
 - A. Develop GRU using a Single Store and a Single Item
 - B. Add Holiday, Day of Week, and Promotions Feature
 - C. Increase to Multi-Store and Multi-Item Prediction
 - D. Develop Order Suggestion and Waste Prediction
 3. LSTM
- V. Performance Evaluation
- VI. Conclusion
- VII. References
- VIII. SELF-DECLARATION

Self-Declaration: We, the undersigned, confirm that this proposal is a group project. Each member contributed to the planning and implementation strategies.